Type equivalence is an equivalence relation: identity, inversion, and composition of equivalences

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled thm:equiv-eqrel (source lines 1169-1177); the proof span 1178-1186 is the author's proof environment, retained line for line.

Editorial note. The retained context carries the author's definition of quasi-inverse; the equivalence-type notation used by the statement is covered by the standard prerequisites (the author defines it immediately before the lemma and the proof does not depend on its internals).

Editorial note. The statement and proof cite no external labels, so no external references are declared.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the equivalence relation laws for type equivalence. Only the counted lemma and its author proof are retained; the author's definition of equivalence and of quasi-inverse is background or retained context as indicated.

Allowed standard prerequisites (editorial).

- The notion of quasi-inverse of a function: a triple consisting of a function in the reverse direction and two homotopies, defined by the retained context.
- The notion of equivalence of types as an improved, structure-respecting predicate on functions, written f in the equivalence type from A to B; the author's definition and its motivation are background for this card.
- Homotopies between functions and identity functions on types.

Definition 0.0.1. For a function $f : A \to B$, a **quasi-inverse** of f is a triple (g, α, β) consisting of a function $g : B \to A$ and homotopies $\alpha : f \circ g \sim \mathsf{id}_B$ and $\beta : g \circ f \sim \mathsf{id}_A$.

Lemma 0.0.2. *Type equivalence is an equivalence relation on $\mathcal{U}$. More specifically:*

(i) For any A, the identity function id_A is an equivalence; hence $A \simeq A$.
(ii) For any $f : A \simeq B$, we have an equivalence $f^{-1} : B \simeq A$.
(iii) For any $f : A \simeq B$ and $g : B \simeq C$, we have $g \circ f : A \simeq C$.

Proof. The identity function is clearly its own quasi-inverse; hence it is an equivalence.

If $f : A \to B$ is an equivalence, then it has a quasi-inverse, say $f^{-1} : B \to A$. Then f is also a quasi-inverse of f^{-1}, so f^{-1} is an equivalence $B \to A$.

Finally, given $f : A \simeq B$ and $g : B \simeq C$ with quasi-inverses f^{-1} and g^{-1}, say, then for any $a : A$ we have $f^{-1}g^{-1}gfa = f^{-1}fa = a$, and for any $c : C$ we have $gff^{-1}g^{-1}c = gg^{-1}c = c$. Thus $f^{-1} \circ g^{-1}$ is a quasi-inverse to $g \circ f$, hence the latter is an equivalence. □